Weighted Fault Tolerant Matroid Spanning with Safe and Vulnerable Elements

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Abstract

Let M be a rank-r matroid given by an independence oracle, let its elements have nonnegative binary-encoded weights, and let V be a specified set of vulnerable elements. We seek a minimum-weight set B that spans M after the deletion of any at most k elements of $B \cap V$. We prove fixed-parameter tractability with parameter $r + k$. The algorithm constructs a subset of size depending only on r, k that contains an optimum solution. Its proof combines a bounded-cardinality optimum, a weight-respecting exchange against deficient hyperplanes, and a recursive construction using uniform sets. Retaining lower-rank branches after reaching a uniform-set threshold is essential for preserving weight. The all-vulnerable specialization gives the weighted extension asked for in the conclusion of Bentert, Fomin, Golovach and Morelle (ESA 2025). We separately explain a notation mismatch in that paper's nonuniform formulation and state precisely the selected-set model treated here. We also give a direct application to robust dedicated-input placement on strongly connected state digraphs.

1 Introduction

Fault-tolerant spanning asks for a set of elements that continues to span a matroid when some selected elements fail. Unlike an ordinary basis, a feasible purchased set may contain redundant elements. With weights, the objective is the total cost of all purchased elements, rather than the cost of a basis chosen after a failure.

Bentert, Fomin, Golovach and Morelle [2] establish fixed-parameter tractability for the unweighted problem parameterized jointly by the rank and failure budget. Their conclusion asks whether this algorithmic result extends to weights and to safe and vulnerable elements. We prove a weighted theorem in the independence-oracle model. No linear representation is required.

Definition 1. Let $M = (E, \mathcal{I})$ have rank r, let $V \subseteq E$, and let $k \geq 0$ be an integer. A set $B \subseteq E$ is *resilient* if

$$\operatorname{rank}_M(B \setminus F) = r \qquad \text{for every } F \subseteq B \cap V \text{ with } |F| \leq k. \tag{1}$$

Elements in $E \setminus V$ are safe. For weights $w : E \to \mathbb{Z}_{\geq 0}$, write $w(B) = \sum_{e \in B} w(e)$. The optimization problem is to minimize $w(B)$ over resilient sets.

Failures could equivalently be quantified over all subsets of V of size at most k: failures outside B have no effect. In either formulation the surviving *selected set*, not the surviving full ground set, must have rank r.

Theorem 2. *For an n-element rank-r matroid given by an independence oracle, nonnegative binary-encoded integer weights, a vulnerable subset V, and an integer $k \geq 0$, a minimum-weight resilient set can be found, or infeasibility reported, in $f(r,k)\,\mathrm{poly}(n,L)$ time and independence-oracle calls, where L is the input bit length and oracle calls have unit cost. For $r \geq 2$, putting $q = (k+1)r$, a bound is*

$$2^{O(qr^2(r+k))}\,\mathrm{poly}(n,L).$$

In particular, this is $2^{O((r+k)^5)}\,\mathrm{poly}(n,L)$. For fixed $k \geq 1$, a bound is $r^{O_k(r^3)}\,\mathrm{poly}(n,L)$. The cases $r \leq 1$ or $k = 0$ are polynomial-time solvable.

The algorithm retains an optimum in an FPT-sized subset of E. This is an oracle reduction with parameter-bounded output cardinality; it is not a claim of a polynomial-size encoded kernel. Oracle access to the original matroid is retained.

1.1 Relationship to previous formulations

With $V = E$, Definition 1 is precisely the weighted fault-tolerant basis problem in Section 1.1 of [2]. Thus the specialization of Theorem 2 proves the weighted FPT statement posed in that paper. The theorem extends the uniform-set decomposition underlying its unweighted algorithm. The additional requirement is to preserve optimum purchase weight throughout the decomposition, including when a large uniform set has been found.

The nonuniform question in Section 7 of both the proceedings and arXiv version of [2] instead prints $\mathrm{rank}(M - F) = \mathrm{rank}(M)$. Taken literally, this condition is not equivalent to (1). In the rank-one uniform matroid $U_{1,2}$ on $\{a,b\}$, let both elements be vulnerable, let $k = 1$, and take $B = \{a\}$. The set B initially spans, and $\mathrm{rank}(M - F) = 1$ for every permitted $F \subseteq B$, but $B \setminus \{a\}$ does not span. Without an initial-spanning requirement, $B = \varnothing$ satisfies the literal condition. Our safe/vulnerable theorem concerns the explicitly stated selected-set interpretation, which agrees with the claimed all-vulnerable special case and the selected-subgraph model of Adjiashvili et al. [3]. We do not assume an author-confirmed correction of the printed formulation.

2 Matroid preliminaries and a small optimum

All matroids in this paper are finite. The rank of $X \subseteq E$ is the maximum size of an independent subset of X. Its closure is

$$\mathrm{cl}(X) = \{e \in E : \mathrm{rank}(X \cup \{e\}) = \mathrm{rank}(X)\}.$$

We use monotonicity and idempotence of closure, and the fact that a maximal independent subset of X spans X. These properties apply to arbitrary matroids. A hyperplane is a flat of rank $r - 1$. Ranks can be computed using at most $|X|$ greedy independence queries.

Loops can be discarded. Removing them from a resilient set preserves resilience, and their nonnegative weights cannot make this removal costly. If $r = 0$, the empty set is optimal. Henceforth assume $r \geq 1$ and work on the loop-free restriction. Set

$$q = (k+1)r, \qquad H = \binom{r+k-1}{k}, \qquad t_h = (h-1)H + q \quad (1 \leq h \leq r). \tag{2}$$

The same global values of q, H are used in every recursive subproblem.

Lemma 3. *Every resilient set B contains a resilient subset B_0 of cardinality at most q. Consequently, if a feasible solution exists, there is a minimum-weight solution of cardinality at most q.*

Proof. Successively choose a maximal independent subset I_j of $B \setminus \bigcup_{i<j} I_i$, for $j = 0, \dots, k$, and put $B_0 = \bigcup_{j=0}^{k} I_j$. Empty layers are allowed. Since $|I_j| \le r$, we have $|B_0| \le q$.

Fix $F \subseteq B_0 \cap V$ with $|F| \le k$. The layers are disjoint, so some I_j avoids F. Every element of $B \setminus B_0$ was available when I_j was chosen and belongs to $\mathrm{cl}(I_j)$. Every element of $B_0 \setminus F$ already lies in that surviving set. Therefore

$$B \setminus F \subseteq \mathrm{cl}(B_0 \setminus F).$$

Since F is also an allowed failure of B, its left-hand side spans M. Thus B_0 is resilient. Nonnegative weights give $w(B_0) \le w(B)$, and applying this construction to an optimum proves the second assertion. □

3 A weight respecting exchange

We first sharpen the deficient-hyperplane count using the classical set-pairs inequality of Bollobás [1]. The short proof is included; no representation or bound on the size of Y is needed.

Lemma 4. *For any $Y \subseteq E$, at most $\binom{r+k-1}{k}$ distinct hyperplanes can have the form $\mathrm{cl}(Y \setminus F)$ with $F \subseteq Y$ and $|F| \le k$. This count is sharp.*

Proof. Index these hyperplanes by P_i. Choose a basis $A_i \subseteq Y \cap P_i$ of size $r-1$, and put $D_i = Y \setminus P_i$. Because $P_i = \mathrm{cl}(Y \setminus F)$ for some permitted F, we have $D_i \subseteq F$, so $|D_i| \le k$. Also $A_i \cap D_i = \varnothing$. If $i \ne j$, then $A_i \cap D_j \ne \varnothing$: otherwise $A_i \subseteq P_j$ and hence $P_i = \mathrm{cl}(A_i) \subseteq P_j$, forcing equality of the two hyperplanes.

In a uniformly random permutation of Y, let $\mathcal{E}_i$ be the event that every element of A_i precedes every element of D_i. Its probability is $\binom{r-1+|D_i|}{r-1}^{-1}$. These events are pairwise disjoint. Indeed, for $i \ne j$ choose $a \in A_i \cap D_j$ and $b \in A_j \cap D_i$. The two events require a before b and b before a, respectively. Therefore

$$\sum_i \binom{r-1+|D_i|}{r-1}^{-1} \le 1.$$

Each summand is at least $\binom{r+k-1}{r-1}^{-1}$, proving the bound. The argument includes $r = 1$ and $k = 0$, when there can be at most one such hyperplane.

For sharpness take $M = U_{r,r+k}$, $B = E$, all elements vulnerable, and $Y = B \setminus \{x\}$. The set B is resilient, and each k-subset of Y leaves a distinct $(r-1)$-element hyperplane. Thus equality holds even for the deficient hyperplanes needed below. This does not assert sharpness of the full core threshold t_h. □

Call Z *h-uniform* if $\mathrm{rank}(Z) = h$ and every h-element subset of Z is independent. Such a set has at least h elements, and every subset of size at most h is independent. A set has a single failure status if it lies entirely in V or entirely in $E \setminus V$.

Lemma 5. *Let $A \subseteq E$ have a single failure status and rank $h \ge 1$. Let $Z \subseteq A$ be h-uniform with $|Z| \ge t_h$. Suppose B is resilient, $|B| \le q$, and $x \in (B \cap A) \setminus Z$ satisfies $w(z) \le w(x)$ for every $z \in Z$. Then some $y \in Z \setminus B$ makes*

$$B' = (B \setminus \{x\}) \cup \{y\}$$

resilient, with $|B'| = |B|$ and $w(B') \le w(B)$.

Proof. Put $Y = B \setminus \{x\}$. Consider the permitted failures $F \subseteq Y \cap V$, $|F| \le k$, for which $\operatorname{rank}(Y \setminus F) < r$. Resilience of B implies that their surviving ranks equal $r-1$ and that $x \notin \operatorname{cl}(Y \setminus F)$. Lemma 4 bounds the number of these distinct hyperplanes by H. It applies also when x is safe and some deficient failure sets have size less than k.

Each of these hyperplanes contains at most $h-1$ elements of Z. Otherwise it contains an independent h-subset of Z, which spans A, and hence contains x, a contradiction. At most $q-1$ elements of Z belong to B, because $x \notin Z$. The number of forbidden choices in Z is at most $(h-1)H + q - 1 < t_h$. Choose a $y \in Z \setminus B$ outside all the deficient hyperplanes.

Let $F' \subseteq B' \cap V$ have size at most k. If $y \notin F'$, then $F' \subseteq Y$; either $Y \setminus F'$ already spans or y repairs its rank deficiency by construction. If $y \in F'$, then y is vulnerable, so x is vulnerable as well. The set

$$(F' \setminus \{y\}) \cup \{x\}$$

is an allowed failure of B of the same size as F'. Its survivor is exactly $B' \setminus F'$, which therefore spans. The cardinality and weight assertions follow from the choice of y. □

The argument includes $r = 1$, when a deficient hyperplane has rank zero and its basis is empty. It does not assume distinct weights, positive weights, or representability.

4 Recursive construction of an optimum preserving core

For a loop-free set A of a single failure status, define $\operatorname{Core}(A)$ by the following procedure. Every scan uses a fixed total ordering that refines nondecreasing weight.

1. If $A = \varnothing$, return $\varnothing$. Let $h = \operatorname{rank}(A)$. If $h = 1$, return the q cheapest elements of A, or all of A if it has fewer. If $|A| \le t_h$, return A.

2. Scan the elements of A in order, initially with $Z = \varnothing$. Accept an element x precisely when every subset of $Z \cup \{x\}$ of size at most h is independent. Stop when $|Z| = t_h$ or all elements have been scanned.

3. For every $T \in \binom{Z}{h-1}$, put $A_T = A \cap \operatorname{cl}(T)$ and recursively compute $\operatorname{Core}(A_T)$. Return

$$Z \cup \bigcup_{T \in \binom{Z}{h-1}} \operatorname{Core}(A_T).$$

In particular, step 3 is executed under *both* stopping conditions. Recursive domains need not be flats of the original matroid.

Until Z has size h, the acceptance rule is ordinary greedy independence. The final Z therefore has at least h elements and is h-uniform. Each child has rank exactly $h-1$: it contains T and is contained in $\operatorname{cl}(T)$.

Lemma 6. *For every resilient B with $|B| \le q$ and every single-status loop-free set A, there is a resilient B' such that*

$$|B'| = |B|, \quad w(B') \le w(B), \quad B' \setminus A = B \setminus A, \quad B' \cap A \subseteq \operatorname{Core}(A).$$

Proof. Induct on $h = \operatorname{rank}(A)$. Empty domains and a return of all of A require no change. If $h = 1$ and $|A| > q$, an element x of $B \cap A$ outside the q cheapest elements can be replaced by an unused retained element y of no larger weight. Nonloop elements of a rank-one domain are parallel. When

y survives, substitution preserves rank; when y fails, it and x are both vulnerable, and the failure mapping in Lemma 5 applies. Repeating gives the claim. Equivalently, use Lemma 5 with $h = 1$ and $|Z| = q$.

Suppose $h > 1$ and the scan is executed. Let $U = \bigcup_T A_T$. Every rejected scanned element x belongs to U. Indeed, at rejection a dependent subset of $Z \cup \{x\}$ contains x and at most $h - 1$ retained elements. Those retained elements are independent, so they span x. Extend them to an $(h - 1)$-subset of the final Z, obtaining a child containing x. Also $Z \subseteq U$, since $h > 1$ and every element of Z belongs to an $(h - 1)$-subset of Z.

If the scan is exhausted, $A = U$. Otherwise each element of $A \setminus U$ is unscanned and is at least as expensive as every element of Z. Apply Lemma 5 successively to the elements of $B \cap (A \setminus U)$. Each replacement lies in $Z \subseteq U$ and removes one selected element outside U. The resulting resilient set has its intersection with A inside U, unchanged cardinality and outside-A part, and no larger weight.

Fix the final output $W_A = \mathrm{Core}(A)$ and process the children $A_1, \ldots, A_m$ in any order. Apply the induction hypothesis to each child and the current solution. For the current solution define $D(B) = (B \cap A) \setminus W_A$. Processing A_i changes only elements in A_i and leaves its selected elements in $\mathrm{Core}(A_i) \subseteq W_A$. Consequently,

$$D(B_{\mathrm{new}}) \subseteq D(B_{\mathrm{old}}) \setminus A_i.$$

Initially $D(B) \subseteq U$. After all children have been processed it is empty, regardless of overlaps between children. All operations preserve cardinality, hence the bound q, and preserve resilience without increasing weight. This establishes the induction. □

Apply the construction separately to the vulnerable and safe nonloop elements, and let W be the union of the two outputs. Their domains are disjoint. Lemmas 3 and 6 imply that W contains an optimum of cardinality at most q whenever the instance is feasible.

5 Size and running time

Put $K_0 = 0$, $K_1 = q$, and, for $2 \le h \le r$, define

$$K_h = t_h + \binom{t_h}{h-1} K_{h-1}.$$

By induction, a rank-h call returns at most K_h elements. The sequence is nondecreasing; in particular $|W| \le 2K_r$, or $|W| \le K_r$ if all elements are vulnerable.

For a convenient explicit bound put $T = t_r$ and $N = 2rT^{(r-1)^2}$. Each of the two recursion trees has at most r levels and branching at most T^{r-1}. Thus there are at most N nodes and $|W| \le NT$. At a node, before h elements are accepted, one independence query tests a candidate. Subsequently it suffices to test the h-subsets containing the candidate. This uses at most $O(n(1 + T^{r-1}))$ independence queries per node. Testing $x \in \mathrm{cl}(T')$ for a child-defining independent $(h - 1)$-set T' uses one independence query to $T' \cup \{x\}$, with the membership case $x \in T'$ handled directly. Computing a domain's rank requires at most n queries. Sorting and set bookkeeping have polynomial input cost multiplied by a parameter-only factor.

Finally enumerate all $B \subseteq W$ with $|B| \le q$, and for each enumerate all allowed failure sets and compute its surviving ranks greedily. There are at most $(NT + 1)^q$ candidates and at most 2^q failures per candidate. Choose a feasible candidate of minimum total weight. If there is none,

optimum preservation proves infeasibility. Exact weight sums and comparisons have polynomial bit cost in the original weight encoding.

For $r \geq 2$, put $p = r + k$. Since $H \leq 2^{p-1}$ and $q \leq (p+1)^2$, we have $\log_2 T = O(p)$. Hence $NT = 2^{O(r^2 p)}$, and enumerating the candidates and failures gives $2^{O(qr^2 p)} \operatorname{poly}(n, L)$. As $q = O(p^2)$, this is at most $2^{O(p^5)} \operatorname{poly}(n, L)$. For fixed $k \geq 1$, the sharper estimate $H = O_k(r^k)$ gives $T = O_k(r^{k+1})$, $NT = r^{O_k(r^2)}$, and running time $r^{O_k(r^3)} \operatorname{poly}(n, L)$. There is no enumeration with exponent r on n, and no pseudopolynomial dependence on weight values.

If $k = 0$, ordinary minimum-weight-basis greedy is optimal. If $r = 1$, an optimum consists of either one cheapest safe nonloop element or $k + 1$ cheapest vulnerable nonloop elements, choosing the cheaper available option. These observations also handle large k without a special feasibility assumption. Nonnegative rational weights can be handled by exact comparisons and sums with polynomial bit complexity; the accompanying implementation accepts integers and exposes a rank-oracle interface, which is simulated from an independence oracle by the greedy procedure described above.

6 Why simpler weighted reductions do not suffice

6.1 Stopping at a large uniform set

Consider a rank-two truncation of a partition matroid with two parallel classes of size two having weights $(0, 0)$ and $(1, 1)$, and five further singleton classes each of weight 100. Let $k = 1$. Then $q = 4$, $H = 2$, and $t_2 = 6$. The weight-ordered uniform scan reaches the threshold while retaining only one element from each cheap parallel class. Its robust optimum is 101, whereas the full instance has optimum 2, attained by the two cheap pairs. Thus sorting the scan is insufficient if a successful threshold causes the recursion to stop. Our construction retains the cheap rejected elements through the lower-rank children.

A rank-three version consists of three parallel pairs of costs $(0, 1)$ and 79 singleton classes of costs $13, \ldots, 91$. For $k = 1$, $H = 3$ and $t_3 = 12$. The uniform-only output has robust optimum 13, while the original optimum is 3. The recursive core retains 15 elements and preserves that optimum. The earlier basis-subset count $\binom{q-1}{r-1}$ gave threshold 26 and a 29-element core on the same instance.

6.2 Preserving cheapest ordinary bases after failures

Dey and Kavitha [5] study a preserver $S \subseteq E$ such that, for every $F \subseteq E$ with $|F| \leq k$, $M|(S \setminus F)$ contains a minimum-cost ordinary basis of $M|(E \setminus F)$. That property need not preserve the optimum total cost of a resilient purchased set.

For a five-element example, take a rank-two truncation with parallel classes A, B of size two and a singleton class C. Give each element of $A \cup B$ weight 2 and the element of C weight 3. For every zero or one ground-element failure, $A \cup B$ still contains an ordinary basis of cost 4, which is minimum in the full surviving matroid. Yet the robust optimum inside $A \cup B$ is 8, and the full instance has optimum 7, using one element from each class. Two successive cheapest residual bases also produce $A \cup B$. Thus this preserver construction does not replace the optimum-preserving recursion above.

7 An application to dedicated input placement

Consider a nonempty strongly connected state digraph. Each selected vertex receives one dedicated actuator, with no repeated actuators at a vertex; selected vulnerable actuators may fail, while state

vertices and state edges remain intact. Let T_0 be the transversal matroid on right-state vertices whose independent sets can be matched using state edges. Write $n = |E|$, $\nu = \operatorname{rank}(T_0)$, and $d = n - \nu$. Its dual $M = T_0^*$ has rank d.

For surviving actuator vertices B, the usual dedicated-input structural-controllability criterion [7] requires a maximum state matching leaving unmatched vertices only in B, together with accessibility. The matching requirement is that $E \setminus B$ be independent in T_0: such an independent set extends to a basis of T_0, which is saturated by a maximum state matching, and the converse is immediate. By the dual rank identity,

$$\operatorname{rank}_M(B) = |B| - \nu + \operatorname{rank}_{T_0}(E \setminus B),$$

this is equivalent to $\operatorname{rank}_M(B) = d$. The independence oracle for M is polynomial-time implementable using bipartite matching and the same identity.

Corollary 7. *Minimum-cost robust dedicated-input placement on a strongly connected state digraph, under the actuator-failure model just specified, is FPT in $d + k$, with nonnegative binary-encoded costs and safe/vulnerable actuator locations.*

Proof. If $d > 0$, every surviving spanning set in M is nonempty. Strong connectivity then ensures accessibility, so Theorem 2 applies. If $d = 0$, matching is automatic and accessibility requires a nonempty survivor. Choose the cheaper of one cheapest safe actuator and $k + 1$ cheapest vulnerable actuators, omitting unavailable options. □

For general state digraphs, additional source strongly connected component constraints are necessary. They are not guaranteed to be preserved by our same-status exchange lemma, so no such extension is claimed here.

8 Related work and scope

The unweighted core construction in [2] is the starting point of this proof. In the proceedings numbering, Proposition 6 gives the disjoint-layer cardinality bound, Lemma 8 already uses the basis-subset hyperplane count $\binom{q-1}{r-1}$ and its exchange framework, and Lemma 9 gives the recursive uniform-set core. Those ingredients are not new here. The changes are the weight-order invariant, retaining closure recursion in both stopping cases, and restricting exchanges to the same failure status. Lemma 4 additionally applies the classical inequality [1] to replace that count by $\binom{r+k-1}{k}$; we do not claim a new set-pairs theorem. That paper also provides polynomial-time weighted algorithms for rank at most two (Corollary 12) and for truncations of unit-capacity partition matroids (Proposition 5). Our contribution is the weighted optimum-preservation argument for arbitrary oracle matroids with parameter $r + k$, together with its safe/vulnerable selected-set version.

Bulk-robust matroid optimization [4] uses the purchase-cost objective and explicitly specified failure scenarios. Definition 1 corresponds to the special family of all at most k vulnerable-element failures. Its exact size is $\sum_{i=0}^{\min(k,|V|)} \binom{|V|}{i}$, which is $\Theta(n^k)$ for fixed k in the all-vulnerable case; a polynomial dependence on the list length alone does not give the parameterized bound proved here.

Representative-family methods [6] preserve independent extensions, including weight-respecting variants, and give algebraic constructions for represented linear matroids and separate constructions for uniform matroids. Their hypotheses and extension predicate differ from the arbitrary-oracle, all-failures optimum-preservation property used here. We make no assertion that these methods cannot provide another proof.

The ESA paper and its currently listed arXiv version explicitly pose the weighted FPT question. The public-source review for this manuscript, dated 4 November 2026, did not identify a subsequent

theorem with the same scope. This is not a certification of priority or an exhaustive absence claim. The model distinction in Section 1.1 remains independent of that literature question.

Reproducibility. The accompanying standard-library Python implementation takes an arbitrary rank oracle and implements the general core-and-enumeration algorithm. It does not include the optional polynomial-time shortcuts for $r = 1$ and $k = 0$ described above; its running time on those cases is therefore not asserted to be polynomial. Separate finite audits use exhaustive subset/failure enumeration and an independent worst-deletion dynamic program. They include nonrepresentable, graphic and sparse-paving examples, zero and tied weights, mixed failure statuses, infeasibility, and both threshold counterexamples above. The five-element preserver distinction is checked independently. These finite checks support implementation validation and do not replace the matroid proofs.

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